

Chimwemwe Kadiwa

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Self-description

Motivated and innovative Biomedical Engineer with hands-on experience in medical device repair, installation, calibration support and maintenance. Underpinned by a strong foundation in Medical Gas Pipeline Systems (MGPS) with practical exposure to cryogenic Liquid Oxygen (LOX) system installation support. Proficient in C++ programming, circuit simulation and prototyping. Passionate about advancing affordable healthcare solutions and contributing to medical innovation, particularly within underserved healthcare systems. A particular interest in strengthening maternal and newborn healthcare through sustainable, practical and context-appropriate technologies.

Work Experience

1. Kamuzu Central Hospital–Physical Asset Management

August 2022 - December 2022

Biomedical Engineering Intern

- Conducted preventative and collective maintenance on medical equipment.
- Carried out decommissioning and safe disposal of non-functional and old model equipment.
- Assisted in procurement processes by preparing technical specifications.
- Conducted risk assessment on hospital equipment and oxygen cylinder to identify potential hazards and improve patient safety.

2. E-Trading/SICO Logistics

May 2025 - Nov 2025

Biomedical engineer

Project: Liquid Oxygen Reticulation system funded by Global Fund under E-Trading/SICO Logistics supervised by CHAI for Mchinji District Hospital, Dedza District Hospital, Kamuzu Central Hospital, Mzimba District Hospital and Rumphi District Hospital.

- Conducted inspection and assessment of hospital facilities in preparation of medical gas pipeline installation.
- Participated in the procurement and installation of medical oxygen piping system.
- Installed and configured automatic oxygen manifold and alarm systems in manifold houses and hospital wards.
- Performed leakage testing and pressure testing in pipes, valves, joints and manifold systems.
- Assisted in system start up, testing, commissioning and operational verification of medical gas systems.

- Provided user training to healthcare personnel on the operation, handling and basic maintenance to nurses, doctors and biomedical engineers.

3. Kamuzu Central Hospital-Physical Asset Management

Biomedical Engineer-Government Internship

September 2025 - June 2026

- Wall mounting of oxygen cylinders
- Performed calibration of medical equipment.
- Installed and commissioned new equipment in various hospital departments.
- Conducted user training of new and old medical equipment.
- Maintained and updated inventory records for existing and newly acquired medical equipment.

4. Innovative Solutions

Biomedical Engineer

January 9 – March 30

Project: Medical Gas piping and Pressure Swing Adsorption (PSA) plant installation.

- Installation and verification of medical gas piping system.
- Installation of automatic manifolds and alarm units in hospital wards
- Participated in installation of oxygen generation plant, medical air system and medical vacuum system.
- Conducted oxygen cylinder filling and monitoring.
- Participated in the maintenance of PSA plant training.

From 24 March 2026 – 30 March 2026, I collaborated with Malawi Bureau of Standards under SICO Logistics Biomedical Engineer. This activity involved the assessment of sites in Mzimba District Hospital, Salima District Hospital, Ntcheu District Hospital and Mulanje District Hospital to verify the integrity and suitability of concrete foundations for Liquid Oxygen tank installation using Ground Penetrating Radar (GPR) and Schmidt hammer.

Education

Malawi University of Business and Applied sciences	2017 - 2023
Bachelor of Science in Biomedical Engineering	

Blantyre Secondary School	2011 - 2014
Malawi School Certificate of Education	

Technical Skills

- **Medical gas pipeline systems installation**
- **Installation of biomedical equipment**
- **Technical reporting and documentation**
- **Oxygen manifolds and cylinder systems installation**

Some of my software skills and soft skills include circuit simulation (Proteus and Multisim), team work and collaboration, leadership, good communication and verbal skills, Microsoft Office suite and programming languages (C++).

Projects

Pulse oximeter calibration device

- The project aimed at developing a low-cost calibration device for pulse

oximeters that would help healthcare professionals in low resource settings. The device was aimed at calibrating pulse oximeters to get accurate readings for better patient outcomes. It focused on accuracy, affordability, and sustainability for neonatal and maternal monitoring by demonstrated practical application of biomedical engineering to improve patient care quality.

Achievements & Awards

-registered with Medical Council of Malawi

Personal Details

- Date of Birth: 20/07/1998
- Nationality: Malawian
- Place: Lilongwe
- Field of work: Biomedical Engineer

Reference

Dr. C. Msosa - Malawi University of Business and Applied sciences

Lecturer

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Mr. Charles Kapalamula - Kamuzu Central Hospital -Physical Asset Management
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Mr. Lameck Chimango – Kamuzu Central Hospital-Physical Asset Management
Senior Biomedical Engineer

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Research Interest

- Engineering solutions for healthcare systems efficiency and accessibility in district hospitals.
- Medical device calibration and standardization for improved healthcare diagnostics.
- Development of low-cost medical devices for low resource healthcare settings.